

Hadamard Gate and Superposition

A Hadamard places a qubit into superposition, typically into either a $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ or $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$



Figure: Circuit from <https://pennylane.ai/qml/glossary/what-is-a-hadamard-gate>



Discrete Fourier Transform

- The Discrete Fourier Transform is a very useful function represented in matrix form by:

$$(F_N)_{j,k} = \frac{\omega_N^{jk}}{\sqrt{N}}$$

- **Key Terms:**

- $N = 2^n$
- n : number of bits used in the representation
- $\omega_N = \exp\left(\frac{2\pi i}{N}\right)$

- **Applications** The Discrete Fourier Transform maps a complex vector $(x_0, \dots, x_N)$ to a complex vector $(y_0, \dots, y_N)$ defined by

$$y_k = \frac{1}{N} \sum_{j=0}^{N-1} x_j \exp\left(\frac{2\pi i j k}{N}\right)$$

- **Time Complexity:** $\mathcal{O}(n2^n)$



Quantum Fourier Transform

- **Quantum Fourier Transform**

- Maps a clear computational basis into a wave with a phase.

- **Inverse Quantum Fourier Transform**

- It finds the phase of a messy superposition and maps it into a computational basis to be measured.

- **Use Cases**

- The period finding problem
 - Quantum Phase Estimation
 - Position to Momentum

- **Limitations**

- Initializing Amplitudes
 - Reading Amplitudes



Circuit Diagram

- The above mathematical calculations motivates the circuit diagram represented below:

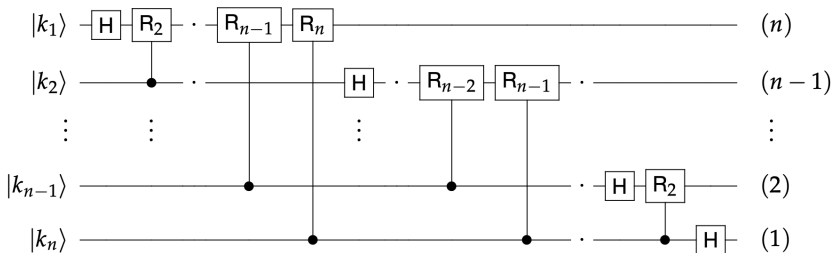


Figure: Circuit from Dalzell, Alexander M. et al.

- Gate Complexity:** $\mathcal{O}(n^2)$ vs FFT $\mathcal{O}(n2^n)$



Quantum Phase Estimation Motivations

- **Pseudo Measurement:** Useful qubit states cannot be measured without collapsing the superposition.
 - Supplies an estimation of a state's phase
 - **Note:** not the same as finding the state
- A lot of useful information can be found in a state's phase
- **Applications:**
 - Quantum Chemistry
 - Shor's
 - Grover's
 - Linear Solvers
- **Note:** Requires a set state via a different algorithm



Why care about Hardware?

- **Fundamental Interest:** We believe quantum information machines can be realized in nature.
- **A "Mathematical Curiosity?":** Without physical realization, the field would be just that.
- **The Challenge:** Building quantum circuits, algorithms, and communication systems is extremely difficult.



Guiding Principles for Physical Realization

- **Qubits:** Robust physical representations of quantum systems.
- **Manipulation:** The ability to control qubits to evolve as desired.
- **Initialization:** The ability to prepare a specific initial state.
- **Measurement:** The ability to read out the final output state.



Which Techniques Are Used Today?

- Not all foundational techniques are commercially pursued for large-scale quantum computers.
- The field is dynamic, with some approaches evolving and others becoming niche research areas.



Techniques That Fell Off

- **Simple Harmonic Oscillator:**
 - Primarily a **theoretical "toy model"**.
 - **Why it fell off:** Not scalable; cannot chain operations; behaves like an analog computer.
- **Nuclear Magnetic Resonance (NMR):**
 - **Why it fell off:** Fundamental scalability issues due to exponential signal loss with more qubits. Fixed molecular architecture limits flexibility.
 - **Current Status:** An excellent educational tool for small-scale quantum algorithm demonstrations, but not for building large commercial quantum computers.
- **Original Optical Photon (using bulk Kerr Media):**
 - **Why it fell off:** Weak interactions and high photon loss in traditional nonlinear optical materials made two-qubit gates impractical.



Techniques That Evolved / Are Used Commercially

- **Trapped Ions:**
 - **How:** Ions are held in electromagnetic traps and manipulated with lasers. This platform has achieved some of the highest gate fidelities.
 - **Companies:** IonQ, Quantinuum, Alpine Quantum Technologies.
- **Cavity QED → Neutral Atoms:**
 - **How:** This is an evolution of the core idea. Arrays of neutral atoms are trapped by optical tweezers and excited to highly-interacting Rydberg states to perform gates.
 - **Companies:** QuEra Computing, Pasqal, Atom Computing.



Other Leading Commercial Platforms

- **Superconducting Qubits:**
 - **How:** Microscopic superconducting circuits cooled to near absolute zero. This is currently the most mature platform for building large processors.
 - **Companies:** Google, IBM, Rigetti.
- **Advanced Photonic Quantum Computing:**
 - **How:** Uses integrated optical circuits to generate, manipulate, and detect photons. Overcomes the limitations of older optical approaches.
 - **Companies:** Xanadu, PsiQuantum.
- **Quantum Dots:**
 - **How:** Uses the spin or charge of electrons confined in nanoscale semiconductor structures. This approach leverages existing semiconductor manufacturing techniques.
 - **Companies:** Intel, Quantum Motion.

